WORKED SOLUTION

STEP III 2014, Question 11

Mechanics • Circular Motion

STEP III 2014, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q11

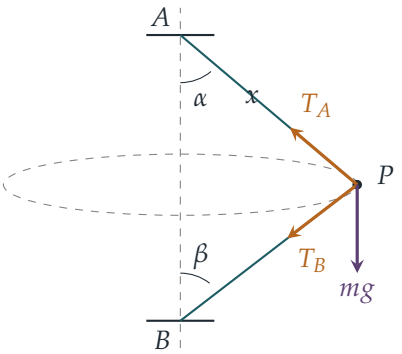
The key idea

Three forces act on the particle (two tensions and its weight), and together they must supply exactly the centripetal force needed for circular motion. Resolving vertically and horizontally gives two linear equations for the two tensions. A string can only pull, so $T_B \geq 0$, and that gives the inequality. In the boundary case the lower string goes slack, and the cosine rule in triangle ABP tells us which angles α can occur.

Solution

Setting up

Model P as a particle and the strings as light and inextensible. P moves in a horizontal circle centred on the line AB, with radius $r = x\sin\alpha$, so its acceleration is $\omega^2 x\sin\alpha$ horizontally towards the axis. The tension T_A acts along PA, making angle α with the upward vertical. The tension T_B acts along PB, making angle β with the downward vertical.



The tensions

Resolving vertically (there is no vertical acceleration) and horizontally towards the axis:

$$T_A \cos\alpha - T_B \cos\beta = mg, \tag{1}$$

$$T_A \sin\alpha + T_B \sin\beta = m\omega^2 x \sin\alpha. \tag{2}$$

These equations still hold if β is obtuse, that is, if P is below B, because then $\cos\beta < 0$ and the vertical component of T_B points upwards, as it should.

To eliminate T_B, multiply (1) by $\sin\beta$ and (2) by $\cos\beta$ and add. Using $\sin\alpha\cos\beta + \cos\alpha\sin\beta = \sin(\alpha+\beta)$,

$$T_A = \frac{m\left(g\sin\beta + \omega^2 x \sin\alpha\cos\beta\right)}{\sin(\alpha+\beta)}.$$

To eliminate T_A, multiply (2) by $\cos\alpha$ and (1) by $\sin\alpha$ and subtract:

$$T_B = \frac{m\sin\alpha\left(\omega^2 x\cos\alpha - g\right)}{\sin(\alpha+\beta)}.$$

Here α and β are two angles of the triangle ABP, so $0 < \alpha+\beta < \pi$ and $\sin(\alpha+\beta) > 0$. Also $\sin\alpha > 0$. A taut string has $T_B \geq 0$, so

$$\omega^2 x\cos\alpha \geq g,$$

as required.

The case $\omega^2 x\cos\alpha = g$

Now $T_B = 0$: the lower string is just taut but carries no tension. Equation (1) then gives

$$T_A = \frac{mg}{\cos\alpha},$$

which agrees with (2), since $T_A = m\omega^2 x$ there. Note also that $\cos\alpha = g/(\omega^2 x) > 0$, so α is acute.

Ask yourself Where does an inequality involving only h, d and α come from?

It comes from the geometry, not the mechanics. Apply the cosine rule to triangle ABP at the vertex A:

$$d^2 = h^2 + x^2 - 2hx\cos\alpha \quad\Longleftrightarrow\quad x^2 - (2h\cos\alpha)\,x + (h^2 - d^2) = 0.$$

The length x is a real root of this quadratic, so its discriminant is non-negative:

$$4h^2\cos^2\alpha - 4(h^2 - d^2) \geq 0 \quad\Longleftrightarrow\quad h^2\cos^2\alpha \geq h^2 - d^2.$$

Both $h\cos\alpha$ and $\sqrt{h^2-d^2}$ are positive (since α is acute and $h > d$), so we may take square roots and get

$$h\cos\alpha \geq \sqrt{h^2 - d^2}.$$

Bounds on T_A

The particle is genuinely off the axis, so $0 < \alpha < \frac{\pi}{2}$ and $0 < \cos\alpha < 1$. Hence $T_A = mg/\cos\alpha > mg$. For the other bound, the previous result gives $\cos\alpha \geq \sqrt{h^2 - d^2}/h$, so

$$mg < T_A = \frac{mg}{\cos\alpha} \leq \frac{mgh}{\sqrt{h^2 - d^2}}.$$

When the upper bound is attained

Equality needs $h\cos\alpha = \sqrt{h^2 - d^2}$, so the discriminant is zero and the quadratic has the repeated root

$$x = h\cos\alpha = \sqrt{h^2 - d^2}.$$

Then $x^2 + d^2 = h^2$, that is, $AP^2 + BP^2 = AB^2$. By the converse of Pythagoras, the angle APB is a right angle: the two strings are perpendicular, and P lies on the circle with diameter AB. Equivalently, AP is tangent to the circle of radius d about B, which is why α is as large as it can be here. In this position $\sin\alpha = d/h$, and P is a height d^2/h above B. For any other acute α allowed by the inequality there are two possible triangles (two values of x), and they merge at this position.

Checks

- If ω is very large, $T_B \approx m\omega^2 x\sin\alpha/\sin(\alpha+\beta)$ grows like ω^2, as both strings must then supply the large centripetal force.
- Take $h = 3$, $d = 2$ at the extreme position: $\cos\alpha = \sqrt{5}/3$, $x = \sqrt{5}$, and $x^2 + d^2 = 5 + 4 = 9 = h^2$. Then $T_A/(mg) = 3/\sqrt{5} \approx 1.342$, equal to $h/\sqrt{h^2 - d^2}$.
- Substituting the formulas for T_A and T_B back into the vector equation of motion, in many random configurations checked by computer, gives zero residual.

Common slips

- Taking the radius of the circle to be x rather than $x\sin\alpha$.
- Giving T_B the wrong vertical sense. With β measured from BA, the string PB pulls P downwards (when β is acute).
- Squaring or square-rooting an inequality without checking that both sides are positive.

Beyond STEP

The boundary case $T_B = 0$ is a constraint becoming *inactive*: a string is a one-sided constraint (it can pull but not push). Problems of this kind, with inequality constraints, lead to Lagrange multipliers with sign conditions in mechanics and optimisation. The

conical pendulum and its relatives are treated well in Kleppner and Kolenkow, *An Introduction to Mechanics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.